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E DELL'INFORMAZIONE

Q-OP

REQUIREMENTS ANALYSIS AND SPECIFICATION DOCUMENT
(RASD)
AUTOMATION AND CONTROL ENGINEERING

Group: CW

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1 | Introduction

1.1. Purpose

The purpose of this project is to apply software engineering practices to develop Q-OP, a web application designed to explicitly manage the usage of specific gym equipment and optimize workout routines.

1.2. Scope

Q-OP addresses the lack of machine-level demand tracking in standard gym booking systems. The system regulates facility access and estimates equipment occupancy by requiring users to book sessions based on their specific workout routines.

1.3. Definitions, Acronyms, Abbreviations

- **RASD:** Requirements Analysis and Specification Document.
- **Routine:** A predefined sequence of gym machines a user intends to use during a session.

1.4. Document Structure

This document outlines the overall description of the system, including scenarios, product functions, and nonfunctional aspects. It also contains the domain model, requirements-level sequence diagrams, and finite state machines.

2 | Overall Description

2.1. Scenarios

Scenario 1: Routine Booking. A user plans to execute a Push/Pull/Legs split. They select their "Push" routine and a Monday 18:00 time slot. The system evaluates the required chest and triceps machines, estimates the occupancy, and displays expected waiting times. The user reviews the prediction and confirms the booking.

Scenario 2: Machine Lifecycle Management. A gym owner notices a cable machine is damaged. They log into the owner dashboard and update the machine's state to "broken". The system updates the machine lifecycle state and immediately invalidates any pending user bookings that require that specific machine, sending notifications to the affected users.

Scenario 3: Routine Optimization. A user attempts to book a routine during a peak hour. The system detects high occupancy for the selected equipment and suggests an alternative time slot or an optimized order for the workout routine to minimize waiting time.

2.2. Domain Model

This class (fig. 2.1) represents the structural foundation of the system, utilizing standard UML notation to map the core entities and their associations. Key relationships include the composition between the Gym and Machine entities, alongside the association between Users and WorkoutRoutines, which encapsulates the central business logic of the booking process.

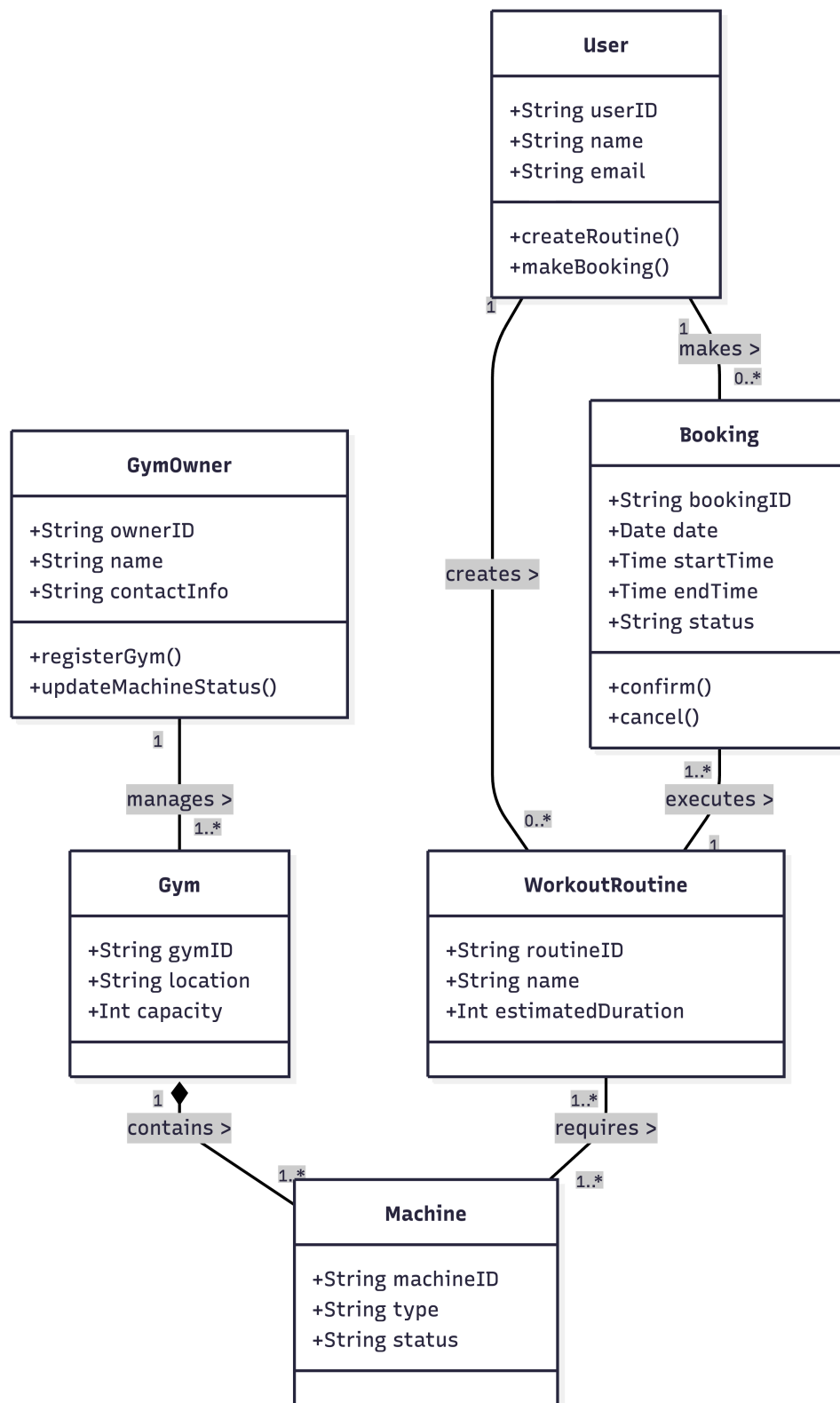


Figure 2.1: Q-OP Domain Model

2.3. User Characteristics

- **Gym Users:** Individuals who follow structured weekly workout routines. They require efficiency and predictability to complete their workouts within limited time windows.
- **Gym Owners:** Facility managers who require a precise view of machine-level demand to make informed decisions regarding equipment maintenance, replacement, and gym capacity.

2.4. Product Functions

- The system shall allow gym owners to register one or more gyms and gym machines.
- The system shall allow users to create and save custom workout routines.
- The system shall calculate and display the expected occupancy of required machines prior to booking confirmation.
- The system shall prevent booking confirmations for unavailable time slots.
- The system shall prevent booking confirmations that involve machines in a broken or occupied state.
- The system shall provide predictive information regarding machine availability and waiting times.
- The system shall suggest alternative time slots, machines, or workout orders when requested equipment is overloaded.

2.5. Nonfunctional Aspects

- **Performance:** The system shall compute and return machine availability predictions and waiting-time estimates within 2 seconds of a user selecting a time slot.
- **Availability:** The booking and notification services shall be available 24/7 to accommodate users planning workouts outside of standard gym operating hours.
- **Scalability:** The system architecture must cleanly separate domain data across multiple gyms without data leakage.

2.6. Assumptions, Dependencies, and Constraints

- **Assumption:** Users physically attend the gym during their confirmed time slots and adhere to the equipment list in their selected workout routines.
- **Assumption:** Gym owners manually and accurately update the lifecycle states of their machines (e.g., from "available" to "under maintenance").

3 | Additional Models

3.1. Requirements-Level Sequence Diagrams

The following models illustrate the dynamic behavior of the Q-OP system through the core functional scenarios.

3.1.1. Routine Booking

This diagram (fig. 3.1) models the booking process, demonstrating the system’s ability to query current machine availability before confirming a reservation. The interaction ensures that capacity constraints are checked before any database write operation occurs.

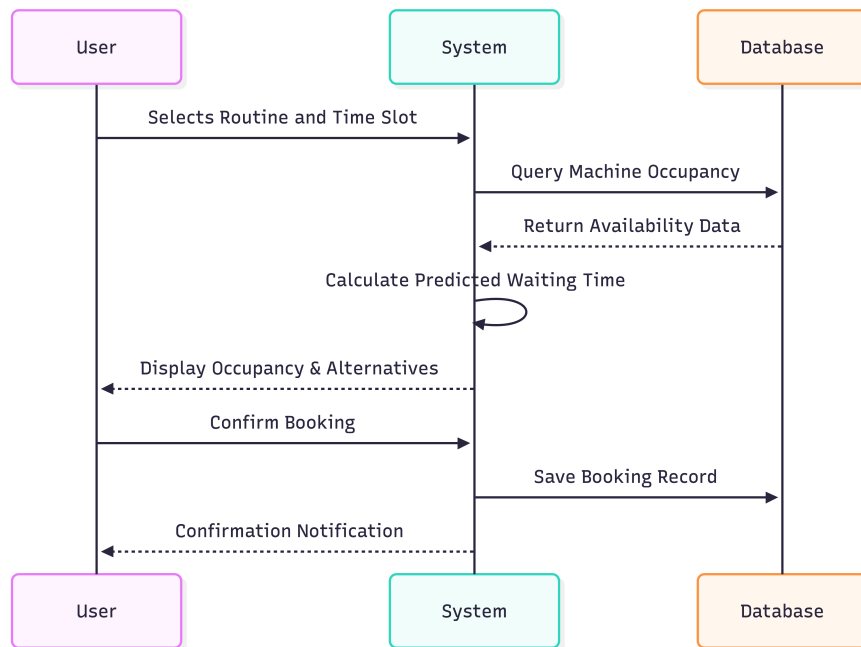


Figure 3.1: Routine Booking Sequence

3.1.2. Machine Lifecycle Management

This diagram (fig. 3.2) shows the process by which a gym owner updates equipment status. The critical requirement modeled here is the system’s automated response—invalidating affected bookings and triggering notifications—to ensure system integrity when a machine is reported as broken.

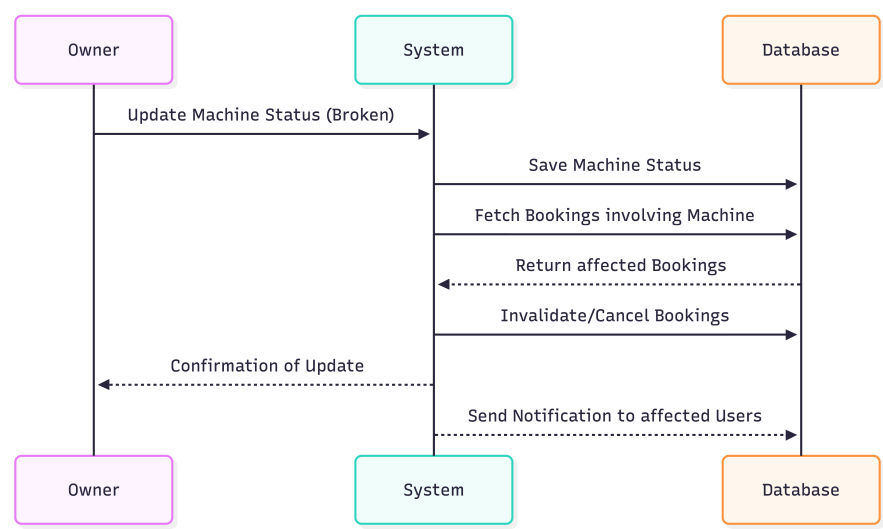


Figure 3.2: Machine Lifecycle Management Sequence

3.1.3. Routine Optimization

This diagram (fig. 3.3) highlights the system’s analytical capabilities. It demonstrates the flow where the system checks real-time occupancy data during peak hours and proposes optimized alternatives to the user, ensuring load distribution across gym resources.

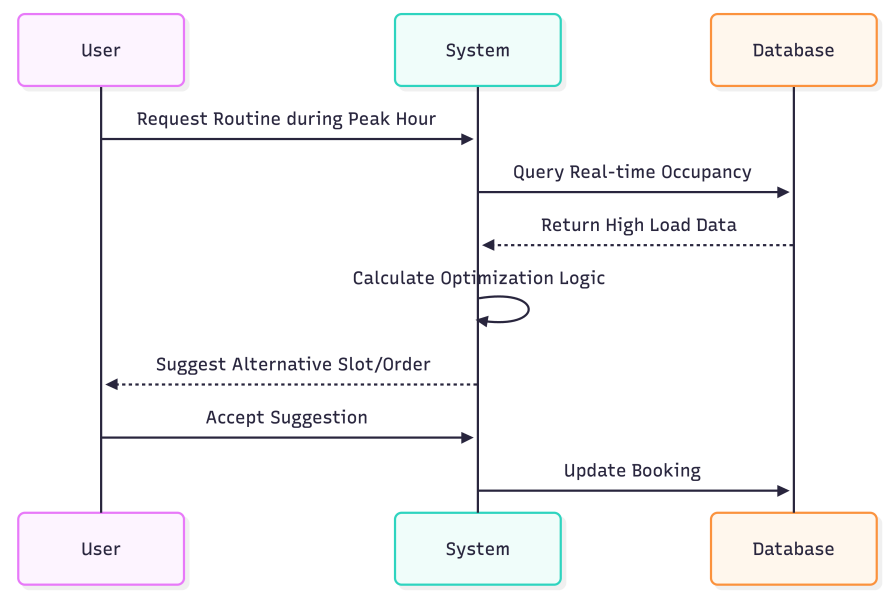


Figure 3.3: Routine Optimization Sequence

3.2. Finite State Machines

This section presents the formal behavioral modeling for the primary operational entities of the Q-OP system: the gym equipment (**Machine**) and the user reservation (**Booking**). These models ensure the system maintains logical consistency across state transitions.

3.2.1. Machine Lifecycle FSM

The Machine Finite State Machine (fig. 3.4) regulates the physical operational status of a piece of equipment. It prevents users from booking resources that are physically unavailable or undergoing service.

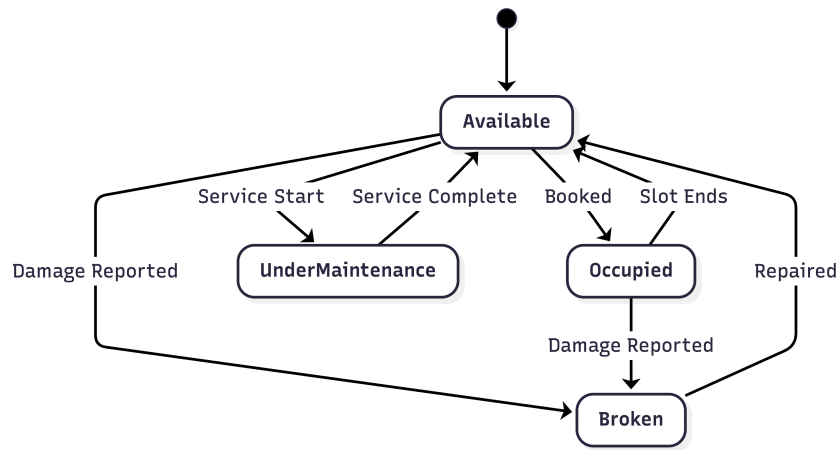


Figure 3.4: Machine Operational Lifecycle FSM

State Descriptions and Transition Logic:

- **Available:** The default initial state of a machine upon registration. The machine is fully functional and open for booking.
- **Occupied:** The machine automatically enters this state when a user's confirmed booking slot starts. It transitions back to *Available* when the slot duration expires.
- **Broken:** Safe failure state triggered manually by a gym owner when damage is reported. This state forces an immediate cancellation of all associated pending bookings. It returns to *Available* only after a successful repair event.
- **Under Maintenance:** A state reserved for scheduled safety checks and preventive servicing.

3.2.2. Booking Lifecycle FSM

The Booking Finite State Machine (fig. 3.5) governs the validity of a user's reservation from initialization to completion or cancellation.

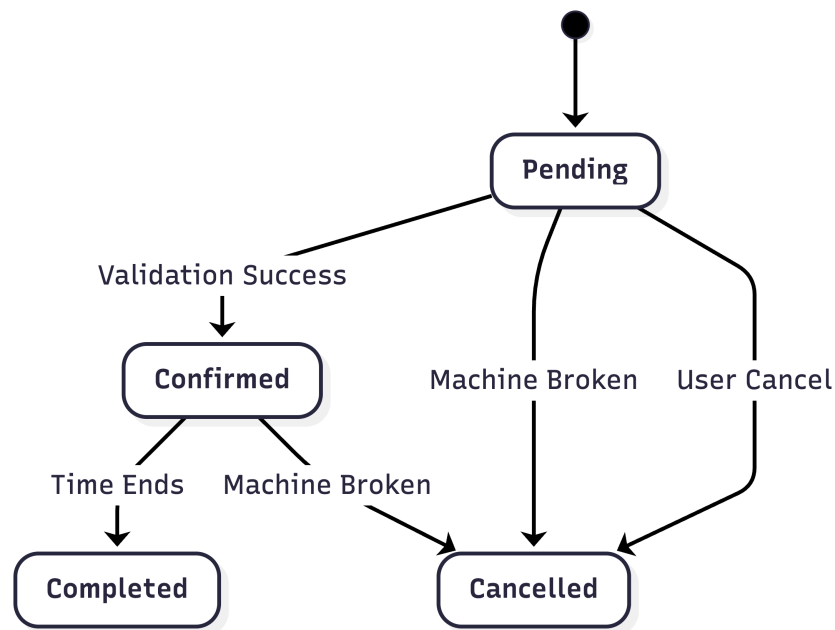


Figure 3.5: User Booking Lifecycle FSM

State Descriptions and Transition Logic:

- **Pending:** The initial state when a user submits a booking request. The system temporarily holds the slot while evaluating machine capacity constraints.
- **Confirmed:** Transited into once the system successfully validates availability. The slot is officially secured for the user.
- **Completed:** The final successful state reached automatically when the allocated time slot concludes.
- **Cancelled:** A terminal state reached either via manual user cancellation or automatically by the system if the target machine transitions into a *Broken* state.

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List of Tables

List of Symbols

Variable	Description	SI unit
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